

3.4.2 P-series

We begin by recalling the Integral Test from last time.

Theorem (The Integral Test): Suppose f is a continuous, positive, decreasing function on $[1, \infty)$ and let $a_n = f(n)$. Then,

1. If $\int_1^{\infty} f(x)dx$ is convergent, then $\sum_{n=1}^{\infty} a_n$ is convergent.
2. If $\int_1^{\infty} f(x)dx$ is divergent, then $\sum_{n=1}^{\infty} a_n$ is divergent.

Example 1 (you try):

- (a) Use the integral test to show that the following series is convergent when the constant, p , satisfies $p > 1$:

$$\sum_{n=1}^{\infty} \frac{1}{n^p}.$$

- (b) Use the integral test to show that the following series is divergent when the constant, p , satisfies

$$0 < p < 1: \sum_{n=1}^{\infty} \frac{1}{n^p}.$$

Theorem (p -series test):

Example 2: Determine whether the following series converges or diverges.

$$1 + \frac{1}{\sqrt[3]{2}} + \frac{1}{\sqrt[3]{3}} + \frac{1}{\sqrt[3]{4}} + \dots$$

Example 3 (you try):

Determine whether the following series converges or diverges: $\sum_{k=1}^{\infty} \frac{9k^{1/2}}{2k^2}$